Projectile Motion of a Marble Shooter

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Abstract

This experiment investigated how launch angle affects the horizontal range of a marble fired from an elevated launcher. A PASCO ME-6825 Mini Launcher was used at its weakest setting to launch the same 16 mm stainless steel marble at twelve angles from 0° to 82.5°, with five trials at each angle. The supplied graph reports mean ranges in meters. The greatest recorded mean range was 1.642 m at 22.5°, compared with 1.412 m at 45°. Complementary angles did not produce equal ranges; for example, 15° and 75° produced mean ranges of 1.587 m and 0.522 m, respectively. Normalizing the measurements by the horizontal-launch range showed that the supplied historical range table did not accurately predict the experimental results. An elevated launch position can explain a maximum below 45° and unequal complementary-angle ranges, although the available measurements do not establish the size of that effect. The results demonstrate the importance of launch conditions when applying projectile-motion models. The exact optimum angle and statistical uncertainty could not be established from the displayed averages alone.

Introduction

Projectile motion describes the movement of objects in two dimensions while traveling through the air. With negligible air resistance, projectile motion consists of horizontal movement with constant velocity and vertical movement with a constant downward acceleration due to gravity. While the two types of movement are independent of one another, they combine to form the characteristic trajectory of a projectile. The initial velocity of the projectile can be separated into horizontal and vertical components according to the initial velocity of the object and its launch angle, θ, and initial speed, v0:

$$v_{0x} = v_0 \cos\theta$$

$$v_{0y} = v_0 \sin\theta$$

The component of acceleration in the horizontal dimension is assumed to be zero; the distance covered horizontally increases linearly with time. The vertical dimension, however, is influenced by gravitational acceleration, approximately 9.81m/s2 near the Earth's surface. These relationships can be represented by the following kinematic equations:

$$x(t) = x_0 + (v_0 \cos\theta)t$$

$$y(t) = y_0 + (v_0 \sin\theta)t - \frac{1}{2}gt^2$$

These equations can be combined to derive the general equation representing the parabolic trajectory of a projectile. With the initial horizontal position defined as zero, this equation is:

$$y(x) = y_0 + x\tan\theta - \frac{gx^2}{2v_0^2\cos^2\theta}$$

In cases where projectiles are both launched and land at the same height, the time of flight and theoretical range of the projectile can be calculated via the following equations:

$$t = \frac{2v_0\sin\theta}{g}$$

$$R = \frac{v_0^2\sin(2\theta)}{g}$$

The theoretical range equation suggests that launch angles of 45 will produce the maximum horizontal displacement, an assumption valid in conditions of constant initial speed and equal heights for both launch and landing. Experimental results may show some divergence from this prediction, however. Factors like air resistance, inconsistencies in mechanical systems, velocities imparted during launch, and measurement uncertainty may all contribute to such differences. The same is true of launching from an elevated position; the additional time available before reaching the landing surface results in a maximum range angle falling below that characteristic of 45. For such cases with an initial height of h above the landing surface, the range is:

$$R = \frac{v_0\cos\theta}{g}\left(v_0\sin\theta + \sqrt{v_0^2\sin^2\theta + 2gh}\right)$$

This equation again considers the difference in heights between launch and landing, an appropriately theoretical model for experiments employing an elevated launcher.

The purpose of the present experiment was to examine the relationship between launch angle and range utilizing a PASCO ME-6825 Mini Launcher and a 16 mm stainless steel marble. Twelve different launch angles between 0 and 82.5 were tested, with five trials performed at each given angle. The resulting distances were analyzed to reveal the angle associated with the greatest range, as well as to compare such findings with theoretical predictions. The study intended not just to evaluate the principles of projectile motion as established by Galileo, but also to note any contributing factors that give rise to differences between theoretical and experimental results.

For the elevated-launch model, h is measured above the landing surface. The positive landing time is

$$t_f = \frac{v_0\sin\theta + \sqrt{v_0^2\sin^2\theta + 2gh}}{g}.$$

For fixed launch height and speed, the ideal angle that maximizes range is

$$\theta_{\text{opt}} = \arctan\left(\frac{v_0}{\sqrt{v_0^2 + 2gh}}\right).$$

When $h > 0$, this angle is below 45°. These equations follow the independent horizontal and vertical motion model [1].

Materials and Methods

The materials used in this experiment included a PASCO ME 6825 Mini Cannon (Sacramento, CA), a 16 mm stainless steel marble (ZY Steel Ball, Amazon), a base plate, clamp, protractor, measuring tape, balance, white paper, and PSLER Blue Carbon Paper for Tracing (A4, 30 pieces). The base plate was secured to the edge of a table using a clamp, and the mini cannon was attached with a bolt, washer, and nut. A pencil was attached to the cannon's string to indicate the launch angle on the protractor.

The marble's mass was measured using a balance, and the horizontal distance from the starting edge of the base plate to the table's edge was recorded. The marble was loaded into the cannon at its first, weakest setting and launched at the designated angle by pulling back and releasing the yellow string. White paper and blue carbon paper were placed in the landing area to mark the marble's impact point, and the horizontal distance traveled was measured in inches.

Five trials were conducted at each of 12 launch angles: 0°, 7.5°, 15°, 22.5°, 30°, 37.5°, 45°, 52.5°, 60°, 67.5°, 75°, and 82.5°, resulting in 60 total trials. The same marble and launcher setting were used throughout the experiment to maintain consistency. After the trials, all equipment was disassembled and returned to its proper location.

Data processing

The mean horizontal range at each angle is calculated by adding the five trial ranges and dividing by five. The analysis uses the recorded mean ranges; the individual trial values are not included in the analyzed dataset. Range values in the data table and graph are expressed in meters; the collection procedure records distances in inches. The conversion is R(m) = 0.0254 × R(in).

The twelve experimental angles described in the methods stop at 82.5°. The plotted dataset also contains the point (90°, 0), whose experimental status is unspecified. This point is treated here as an unverified endpoint rather than an additional set of trials. The ideal vertical-launch model predicts zero horizontal range.

For each angle, the sample standard deviation would be $s = \sqrt{\sum_i (R_i - \bar{R})^2/(n-1)}$, with n = 5, and the standard error of the mean would be $s/\sqrt{5}$. Neither quantity can be recovered from the means alone. No numerical error bars are assigned. Launch height, measured marble mass, individual trials, instrument precision, and the reference-point offset mentioned in the methods are not numerically documented in the analyzed dataset, so none are assumed.

Results

The purpose of this experiment was to investigate the relationship between launch angle and horizontal range using a PASCO ME-6825 Mini Launcher and a 16 mm stainless steel ball. Projectile motion consists of two independent components: horizontal motion, which maintains a constant

velocity when air resistance is negligible, and vertical motion, which experiences a constant downward acceleration due to gravity. These relationships can be represented by the equations

$$x(t) = (v_0 \cos\theta)t$$

$$y(t) = y_0 + (v_0 \sin\theta)t - \frac{1}{2}gt^2$$

Together, these equations explain why a projectile follows a parabolic trajectory.

Based on the experimental data, the greatest average horizontal distance occurred at a launch angle of 22.5°, with an average measured distance of 1.642 in the units recorded during the experiment. As the launch angle increased beyond 22.5°, the horizontal range generally decreased, reaching zero at 90°. This differs from the theoretical prediction that the maximum horizontal range occurs at 45°, as described by the equation

$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

However, this equation assumes that the projectile launches and lands at the same height and that its initial speed remains constant. Since the launcher was positioned above the landing surface, these assumptions may not have applied to the experiment. The launcher's mechanical behavior and measurement uncertainties may have also affected the results.

Overall, the experiment demonstrated that launch angle significantly affects horizontal range and that increasing the angle does not necessarily increase the distance traveled. The results also illustrated the importance of considering experimental conditions when comparing measured values with theoretical predictions. Although the maximum range occurred at a lower angle than expected, the experiment helped demonstrate the relationship between horizontal velocity, vertical acceleration, and projectile motion, supporting Galileo's explanation of a curved, parabolic trajectory.

Table 1 Mean ranges and historical predictions

One point is interpreted as 7.5°, so six points corresponds to 45°. The reference range is the measured horizontal-launch mean, R = 1.422 m. The experimental ratio is R/R, and the historical prediction is the supplied multiplier times R. These predictions are a test of the assigned table, not measured distances. The supplied assignment omits nine points, so no historical value is invented for 67.5°.

Angle (°)	Mean (m)	Mean/R	Multiplier	Prediction (m)
0	1.422	1.000	1.0	1.422
7.5	1.570	1.104	2.8	3.982
15	1.587	1.116	4.6	6.541
22.5	1.642	1.155	6.4	9.101
30	1.622	1.141	8.2	11.660
37.5	1.602	1.127	9.5	13.509
45	1.412	0.993	10.0	14.220
52.5	1.396	0.982	9.5	13.509
60	1.014	0.713	8.2	11.660

Angle (°)	Mean (m)	Mean/R	Multiplier	Prediction (m)
67.5	0.922	0.648	Not supplied	—
75	0.522	0.367	4.6	6.541
82.5	0.250	0.176	2.8	3.982

The displayed 90° value is 0 m and the assigned twelve-point multiplier is 0.0; its status as a measured value remains unverified.

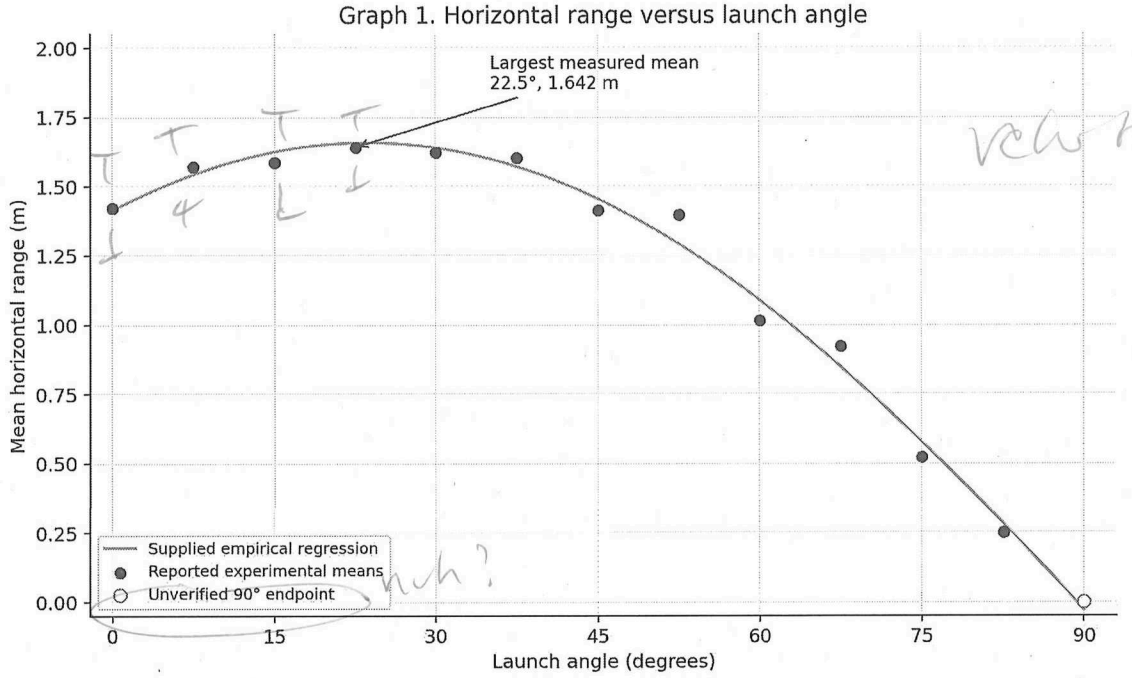


Figure 1: Graph 1 Mean horizontal range versus launch angle

The filled circles show the twelve supplied experimental means, in meters. The open circle at 90° is the unverified endpoint displayed in the original graph. The orange curve reproduces the supplied empirical sinusoidal regression; it is not a calibrated projectile-motion prediction. No error bars are shown because the individual trials are unavailable.

The empirical fit is:

$$R_{\text{fit}}(\theta) = 2.04117 \sin(0.0210174\theta + 1.07579) - 0.383007$$

In this form, is entered as its numerical value in degrees, the sine argument is in radians, and the fitted range is in meters. The fit has $R^2 = 0.9916$ and includes the 90° endpoint. The curve reaches its maximum near 23.55°, while the largest measured mean remains at 22.5°. The fitted maximum is an interpolation, not a measured optimum. A high R^2 describes agreement with the plotted values; it does not independently verify a physical model.

The mean at 22.5° exceeded the mean at 45° by 0.230 m, or approximately 16.3% relative to the 45° mean. However, its advantage over the 30° mean was only 0.020 m. Without trial-level variability,

it is not possible to determine whether that smaller difference is statistically meaningful. The data support identifying 22.5° as the best tested angle, rather than claiming an exact optimum.

Discussion

Tartaglia vs Galileo

The understanding of projectile motion has developed significantly throughout history, particularly through the contributions of Niccolò Tartaglia and Galileo Galilei. Tartaglia proposed that a projectile initially travels along a relatively straight path, gradually transitions into a curved trajectory, and eventually descends almost vertically toward the ground. Although his model represented an early attempt to explain projectile motion, it did not accurately account for the continuous influence of gravitational acceleration throughout the projectile's flight. In comparison, Galileo proposed that projectile motion follows a smooth parabolic trajectory resulting from the combination of constant horizontal velocity and uniform vertical acceleration due to gravity. Galileo's mathematical explanation provides a more accurate representation of projectile motion and establishes the theoretical foundation for analyzing the experimental results obtained in this investigation.

Tartaglia's conceptual representation of projectile motion, illustrating the transition from an initially straight trajectory to a curved path and an almost vertical descent.

Galileo's representation of projectile motion, demonstrating how constant horizontal velocity and uniform downward acceleration combine to produce a parabolic trajectory.

The experimental results were more consistent with Galileo's mathematical model than with Tartaglia's earlier conceptual explanation. According to the data presented in Graph 1, the horizontal range varied with the launch angle, demonstrating the relationship between the horizontal and vertical components of the marble's initial velocity. The greatest average horizontal distance was recorded at a launch angle of 22.5, rather than the theoretically predicted 45 for a projectile launched and landing at the same height. This difference can be explained partly by the experimental setup, in which the marble was launched from an elevated position and landed on a lower surface. Because the projectile had additional time to travel horizontally before reaching the landing surface, the angle producing the greatest horizontal range could have been less than 45.

The theoretical range equation for equal launch and landing heights is:

$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

This equation predicts that the maximum horizontal displacement occurs at 45, since $\sin(90°) = 1$. However, this relationship assumes that the projectile launches and lands at the same vertical position. Because the marble launcher was positioned above the landing surface, the experimental conditions did not fully satisfy this assumption. The more general range equation, which accounts for the vertical height difference, provides a better representation of the experiment. This distinction demonstrates the importance of considering initial conditions when applying theoretical equations to physical systems.

Additionally, five trials were conducted at each of the twelve launch angles, resulting in sixty experimental trials. Repeating the launches helped improve the reliability of the collected data by reducing the influence of random measurement variations on the calculated averages. The same

stainless steel marble and launcher setting were maintained throughout the experiment to minimize inconsistencies in the initial conditions. Nevertheless, several potential sources of experimental error may have influenced the measured horizontal distances. Small variations in the spring-loaded launching mechanism could have produced differences in initial velocity, while uncertainty in identifying the marble's landing position and measuring the horizontal displacement could have affected the recorded values. Air resistance, although considered negligible in the theoretical model, may also have contributed to minor differences between experimental and predicted results.

Another important observation was that increasing the launch angle did not continuously increase the horizontal range. At lower angles, the marble possessed a greater horizontal component of velocity but a smaller upward component, limiting the amount of time available for horizontal travel. As the launch angle increased, the vertical component of velocity became greater, allowing the marble to remain in the air longer while reducing its horizontal velocity. The balance between these two components determines the total horizontal displacement of the projectile. At sufficiently steep launch angles, the reduced horizontal velocity causes the range to decrease despite the increased flight time. This relationship is consistent with Galileo's explanation that projectile motion results from the combination of independent horizontal and vertical components.

Overall, the investigation demonstrated that launch angle significantly influences the horizontal range of a projectile and that the maximum experimental range does not necessarily occur at 45 when the launch and landing heights differ. Although the measured results did not directly verify the ideal maximum-range prediction, they remained consistent with the general principles underlying Galileo's explanation of projectile motion. The experiment also highlighted the importance of accounting for gravitational acceleration, initial velocity, launch height, and experimental uncertainty when comparing measured projectile behavior with theoretical mathematical models. These findings support Galileo's mathematical framework over Tartaglia's earlier conceptual representation while demonstrating how real experimental conditions can influence projectile motion.

Evaluation of the historical range table

The table does not quantitatively match this experiment when R is defined as the measured range at 0°. At 22.5°, it predicts 6.4R = 9.101 m, compared with the measured mean of 1.642 m. At 45°, it predicts 10R = 14.220 m, compared with 1.412 m. The measured maximum is only about 1.155R. Thus, the table overestimates these ranges and places the maximum at the wrong tested angle for this apparatus. This conclusion concerns the supplied table under this normalization; it is not a test of every historical artillery setup. A positive horizontal-launch range also shows why an equal-height model alone cannot describe this elevated-launch experiment: that ideal model gives R(0°) = 0.

In flight trajectory

Under the negligible-drag model, gravity acts throughout the flight. Horizontal velocity remains constant, while vertical velocity decreases during ascent, becomes zero at the highest point, and then becomes downward. The resulting path is a continuous parabola [1, 2]. A range-versus-angle graph is not an image of that flight path: each dot represents a different launch angle and its landing distance. Therefore, these landing measurements alone do not directly demonstrate the trajectory's shape or decisively distinguish historical trajectory models. Video tracking of horizontal and vertical positions over time would provide a more direct test.

Symmetry of complementary angles

The measured ranges are not symmetric about 45°. The complementary pair 15° and 75° gives 1.587 m and 0.522 m, a difference of 1.065 m. Similarly, 22.5° and 67.5° give 1.642 m and 0.922 m, while 30° and 60° give 1.622 m and 1.014 m. Complementary angles have equal ideal ranges only under the equal-height, equal-speed assumptions, because $\sin[2(90° \ )] = \sin(2)$. The elevated-launch formula contains the additional height term and does not require this symmetry. The range asymmetry should not be confused with the geometric symmetry of a complete ideal parabola.

Physical basis of the range model

The model combines constant horizontal velocity with constant downward acceleration. Solving the vertical equation for the positive landing time and multiplying that time by horizontal velocity produces the general range equation. It predicts the tradeoff between increased flight time and decreased horizontal speed as angle increases. The experiment is qualitatively compatible with that explanation, but the measured height and launch speed are needed to calculate an independent theoretical curve. The screenshot setting $v = 10$ is not a measured launch speed. Moreover, at $h = 0$ the general formula must agree with the equal-height formula when both use the same speed, gravitational acceleration, and angle units; separated curves under those identical settings indicate an equation or setting mismatch.

High angle launches and elevated targets

A high-angle launch could be useful for clearing a wall or approaching a target from above, but it reduces horizontal reach at sufficiently steep angles. In this experiment, the mean range fell from 1.642 m at 22.5° to 1.014 m at 60°, 0.522 m at 75°, and 0.250 m at 82.5°. These results show that the greatest available angle is not automatically the best choice. Evaluating an elevated target would require its horizontal distance and height, the launch speed, and the launch height. A suitable trajectory must clear the obstacle and intersect the target; floor-landing ranges alone cannot establish that. The marble experiment also cannot directly provide full-scale cannon ranges.

Limitations and improvements

The launcher height is a plausible contributor to the observed maximum below 45°, but the supplied data do not isolate it from changing launch speed, angle error, or distance-reference error. Any horizontal offset must be applied consistently so the measured range starts beneath the actual launch point. Future work should record muzzle height at each angle, measure launch speed using a photogate or video, retain all individual ranges, and test more angles between 15° and 30°. Repeated measurements would then allow uncertainty estimates and a meaningful comparison of nearby mean ranges. Video tracking would test the in-flight motion independently of the landing-range analysis.

Conclusion

For the tested settings, the largest recorded mean range was 1.642 m at 22.5°. The normalized measurements disagreed with the assigned historical range table and did not show equal complementary-angle ranges. The elevated-launch model provides a physically reasonable explanation for these patterns, but missing launch-condition measurements and trial-level uncertainty prevent a complete quantitative validation.

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References

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